

Market Basket Analysis

```
# Create a list of baskets
market_basket <- list(
  c("bread", "milk"),
  c("bread", "diapers", "beer", "eggs"),
  c("milk", "diapers", "beer", "cola"),
  c("bread", "milk", "diapers", "beer"),
  c("bread", "milk", "diapers", "cola")
)

# set transaction names (T1 to T5)
names(market_basket) <- paste("T", 1:length(market_basket), sep = "")
```

“arules” package and get some outputs

```
library(arules)
```

```
## Loading required package: Matrix
```

```
##
```

```
## Attaching package: 'arules'
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##      abbreviate, write
```

```
library(arulesViz)
```

```
# Transformation
```

```
trans <- as(market_basket, "transactions")
```

```
# Dimensions
```

```
dim(trans)
```

```
## [1] 5 6
```

```
# Item Labels
```

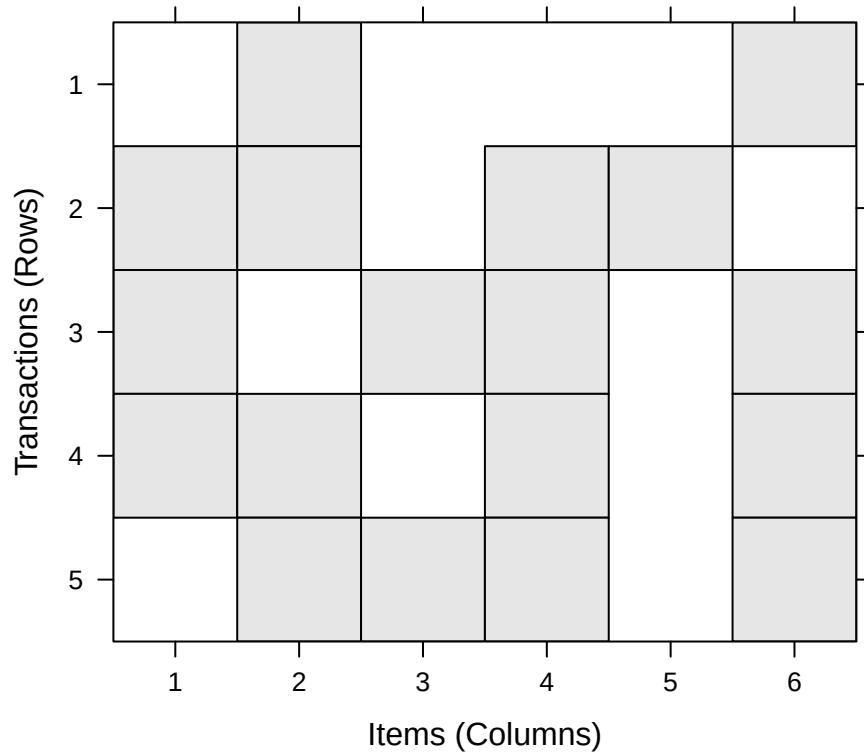
```
itemLabels(trans)
```

```
## [1] "beer"      "bread"     "cola"      "diapers"   "eggs"      "milk"
```

```
# Summary  
summary(trans)
```

```
## transactions as itemMatrix in sparse format with  
## 5 rows (elements/itemsets/transactions) and  
## 6 columns (items) and a density of 0.6  
##  
## most frequent items:  
## bread diapers milk beer cola (Other)  
## 4 4 4 3 2 1  
##  
## element (itemset/transaction) length distribution:  
## sizes  
## 2 4  
## 1 4  
##  
## Min. 1st Qu. Median Mean 3rd Qu. Max.  
## 2.0 4.0 4.0 3.6 4.0 4.0  
##  
## includes extended item information - examples:  
## labels  
## 1 beer  
## 2 bread  
## 3 cola  
##  
## includes extended transaction information - examples:  
## transactionID  
## 1 T1  
## 2 T2  
## 3 T3
```

```
# Plot  
image(trans)
```



```
inspect(trans)
```

```
##      items                      transactionID
## [1] {bread, milk}                T1
## [2] {beer, bread, diapers, eggs} T2
## [3] {beer, cola, diapers, milk}  T3
## [4] {beer, bread, diapers, milk} T4
## [5] {bread, cola, diapers, milk} T5
```

Apriori algorithm in “trans” with minimum support of 0.3 and min. confidence of 0.5

```
rules <- apriori(
  trans,
  parameter = list(supp=0.3,
                    conf=0.5,
                    maxlen=10,
                    target="rules")
)
```

```
## Apriori
##
## Parameter specification:
```

```
## confidence minval smax arem aval originalSupport maxtime support minlen
##      0.5    0.1    1 none FALSE          TRUE      5    0.3    1
## maxlen target  ext
##      10  rules TRUE
##
## Algorithmic control:
## filter tree heap memopt load sort verbose
##      0.1 TRUE TRUE  FALSE TRUE    2    TRUE
##
## Absolute minimum support count: 1
##
## set item appearances ...[0 item(s)] done [0.00s].
## set transactions ...[6 item(s), 5 transaction(s)] done [0.00s].
## sorting and recoding items ... [5 item(s)] done [0.00s].
## creating transaction tree ... done [0.00s].
## checking subsets of size 1 2 3 done [0.00s].
## writing ... [32 rule(s)] done [0.00s].
## creating S4 object ... done [0.00s].
```

```
summary(rules)
```

```
## set of 32 rules
##
## rule length distribution (lhs + rhs):sizes
##  1  2  3
##  4 16 12
##
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1.00   2.00   2.00   2.25   3.00   3.00
##
## summary of quality measures:
##      support      confidence      coverage      lift
## Min.   :0.4000  Min.   :0.5000  Min.   :0.4000  Min.   :0.8333
## 1st Qu.:0.4000  1st Qu.:0.6667  1st Qu.:0.6000  1st Qu.:0.8333
## Median :0.4000  Median :0.7500  Median :0.6000  Median :1.0000
## Mean   :0.4938  Mean   :0.7474  Mean   :0.6813  Mean   :1.0473
## 3rd Qu.:0.6000  3rd Qu.:0.8000  3rd Qu.:0.8000  3rd Qu.:1.2500
## Max.   :0.8000  Max.   :1.0000  Max.   :1.0000  Max.   :1.6667
##      count
## Min.   :2.000
## 1st Qu.:2.000
## Median :2.000
## Mean   :2.469
## 3rd Qu.:3.000
## Max.   :4.000
##
## mining info:
## data ntransactions support confidence
## trans      5      0.3      0.5
##
## apriori(data = trans, parameter = list(supp = 0.3, conf = 0.5, maxlen = 10, target = "rules"))
```

call

```
inspect(rules)
```

##	lhs	rhs	support	confidence	coverage	lift	count
## [1]	{}	=> {beer}	0.6	0.6000000	1.0	1.0000000	3
## [2]	{}	=> {milk}	0.8	0.8000000	1.0	1.0000000	4
## [3]	{}	=> {bread}	0.8	0.8000000	1.0	1.0000000	4
## [4]	{}	=> {diapers}	0.8	0.8000000	1.0	1.0000000	4
## [5]	{cola}	=> {milk}	0.4	1.0000000	0.4	1.2500000	2
## [6]	{milk}	=> {cola}	0.4	0.5000000	0.8	1.2500000	2
## [7]	{cola}	=> {diapers}	0.4	1.0000000	0.4	1.2500000	2
## [8]	{diapers}	=> {cola}	0.4	0.5000000	0.8	1.2500000	2
## [9]	{beer}	=> {milk}	0.4	0.6666667	0.6	0.8333333	2
## [10]	{milk}	=> {beer}	0.4	0.5000000	0.8	0.8333333	2
## [11]	{beer}	=> {bread}	0.4	0.6666667	0.6	0.8333333	2
## [12]	{bread}	=> {beer}	0.4	0.5000000	0.8	0.8333333	2
## [13]	{beer}	=> {diapers}	0.6	1.0000000	0.6	1.2500000	3
## [14]	{diapers}	=> {beer}	0.6	0.7500000	0.8	1.2500000	3
## [15]	{milk}	=> {bread}	0.6	0.7500000	0.8	0.9375000	3
## [16]	{bread}	=> {milk}	0.6	0.7500000	0.8	0.9375000	3
## [17]	{milk}	=> {diapers}	0.6	0.7500000	0.8	0.9375000	3
## [18]	{diapers}	=> {milk}	0.6	0.7500000	0.8	0.9375000	3
## [19]	{bread}	=> {diapers}	0.6	0.7500000	0.8	0.9375000	3
## [20]	{diapers}	=> {bread}	0.6	0.7500000	0.8	0.9375000	3
## [21]	{cola, milk}	=> {diapers}	0.4	1.0000000	0.4	1.2500000	2
## [22]	{cola, diapers}	=> {milk}	0.4	1.0000000	0.4	1.2500000	2
## [23]	{diapers, milk}	=> {cola}	0.4	0.6666667	0.6	1.6666667	2
## [24]	{beer, milk}	=> {diapers}	0.4	1.0000000	0.4	1.2500000	2
## [25]	{beer, diapers}	=> {milk}	0.4	0.6666667	0.6	0.8333333	2
## [26]	{diapers, milk}	=> {beer}	0.4	0.6666667	0.6	1.1111111	2
## [27]	{beer, bread}	=> {diapers}	0.4	1.0000000	0.4	1.2500000	2
## [28]	{beer, diapers}	=> {bread}	0.4	0.6666667	0.6	0.8333333	2
## [29]	{bread, diapers}	=> {beer}	0.4	0.6666667	0.6	1.1111111	2
## [30]	{bread, milk}	=> {diapers}	0.4	0.6666667	0.6	0.8333333	2
## [31]	{diapers, milk}	=> {bread}	0.4	0.6666667	0.6	0.8333333	2
## [32]	{bread, diapers}	=> {milk}	0.4	0.6666667	0.6	0.8333333	2

```
rules <- apriori(
  trans,
  parameter = list(supp=0.3,
                    conf=0.5,
                    minlen=2,
                    maxlen=10,
                    target="rules")
)
```

```
## Apriori
##
## Parameter specification:
## confidence minval smax arem aval originalSupport maxtime support minlen
##          0.5   0.1   1 none FALSE                TRUE         5    0.3    2
## maxlen target ext
##          10 rules TRUE
##
```

```
## Algorithmic control:
## filter tree heap memopt load sort verbose
## 0.1 TRUE TRUE FALSE TRUE 2 TRUE
##
## Absolute minimum support count: 1
##
## set item appearances ...[0 item(s)] done [0.00s].
## set transactions ...[6 item(s), 5 transaction(s)] done [0.00s].
## sorting and recoding items ... [5 item(s)] done [0.00s].
## creating transaction tree ... done [0.00s].
## checking subsets of size 1 2 3 done [0.00s].
## writing ... [28 rule(s)] done [0.00s].
## creating S4 object ... done [0.00s].
```

```
inspect(rules)
```

	lhs	rhs	support	confidence	coverage	lift	count
## [1]	{cola}	=> {milk}	0.4	1.0000000	0.4	1.2500000	2
## [2]	{milk}	=> {cola}	0.4	0.5000000	0.8	1.2500000	2
## [3]	{cola}	=> {diapers}	0.4	1.0000000	0.4	1.2500000	2
## [4]	{diapers}	=> {cola}	0.4	0.5000000	0.8	1.2500000	2
## [5]	{beer}	=> {milk}	0.4	0.6666667	0.6	0.8333333	2
## [6]	{milk}	=> {beer}	0.4	0.5000000	0.8	0.8333333	2
## [7]	{beer}	=> {bread}	0.4	0.6666667	0.6	0.8333333	2
## [8]	{bread}	=> {beer}	0.4	0.5000000	0.8	0.8333333	2
## [9]	{beer}	=> {diapers}	0.6	1.0000000	0.6	1.2500000	3
## [10]	{diapers}	=> {beer}	0.6	0.7500000	0.8	1.2500000	3
## [11]	{milk}	=> {bread}	0.6	0.7500000	0.8	0.9375000	3
## [12]	{bread}	=> {milk}	0.6	0.7500000	0.8	0.9375000	3
## [13]	{milk}	=> {diapers}	0.6	0.7500000	0.8	0.9375000	3
## [14]	{diapers}	=> {milk}	0.6	0.7500000	0.8	0.9375000	3
## [15]	{bread}	=> {diapers}	0.6	0.7500000	0.8	0.9375000	3
## [16]	{diapers}	=> {bread}	0.6	0.7500000	0.8	0.9375000	3
## [17]	{cola, milk}	=> {diapers}	0.4	1.0000000	0.4	1.2500000	2
## [18]	{cola, diapers}	=> {milk}	0.4	1.0000000	0.4	1.2500000	2
## [19]	{diapers, milk}	=> {cola}	0.4	0.6666667	0.6	1.6666667	2
## [20]	{beer, milk}	=> {diapers}	0.4	1.0000000	0.4	1.2500000	2
## [21]	{beer, diapers}	=> {milk}	0.4	0.6666667	0.6	0.8333333	2
## [22]	{diapers, milk}	=> {beer}	0.4	0.6666667	0.6	1.1111111	2
## [23]	{beer, bread}	=> {diapers}	0.4	1.0000000	0.4	1.2500000	2
## [24]	{beer, diapers}	=> {bread}	0.4	0.6666667	0.6	0.8333333	2
## [25]	{bread, diapers}	=> {beer}	0.4	0.6666667	0.6	1.1111111	2
## [26]	{bread, milk}	=> {diapers}	0.4	0.6666667	0.6	0.8333333	2
## [27]	{diapers, milk}	=> {bread}	0.4	0.6666667	0.6	0.8333333	2
## [28]	{bread, diapers}	=> {milk}	0.4	0.6666667	0.6	0.8333333	2

```
plot(rules, engine="plotly")
```

```
## To reduce overplotting, jitter is added! Use jitter = 0 to prevent jitter.
```